

(โครงร่างวิทยานิพนธ์)
เทคนิคที่มีประสิทธิภาพแบบ QUBO สำหรับการแก้ปัญหาการตัดสูงสุด
ในกราฟขนาดใหญ่

นายกฤษฎา สารวิทย์

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(Thesis Proposal)

EFFICIENT QUBO-BASED TECHNIQUES FOR SOLVING THE
MAXIMUM CUT PROBLEM IN LARGE GRAPHS

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Chapter I

INTRODUCTION

1.1 Motivation

The Maximum Cut Problem (Max-Cut), which seeks to partition a graph's vertices to maximize the weight of edges between two subsets, is a significant challenge in combinatorial optimization with wide-reaching applications. From network design and circuit layout to machine learning and data clustering (Ding et al., 2001), the Max-Cut problem appears in numerous real-world scenarios. Traditional methods, while effective for small to medium-sized graphs, often struggle with scalability in larger graphs, especially as problem size grows. Quantum Unconstrained Binary Optimization (QUBO) presents an opportunity to take advantage of quantum properties such as superposition and entanglement to explore solution spaces more efficiently (Rizvee and Khan, 2023). By developing QUBO-based techniques specifically for large graphs, this research aims to bring enhanced performance to the Max-Cut problem, especially for cases where classical algorithms reach their computational limits.

1.2 Research Statement

The development and application of QUBO-based quantum algorithms to effectively solve the Maximum Cut problem in big graphs is the aim of this study. In contrast to classical approaches, this research aims to increase solution quality and processing efficiency by using quantum qualities and hybrid frameworks that blend classical and quantum processes.

1.3 Contribution

1. We propose a novel QUBO-based approach tailored to the Maximum Cut problem in large graphs, optimizing the formulation to improve computational efficiency.
2. Developing a hybrid quantum-classical algorithm that incorporates quantum subroutines with classical optimization techniques, capitalizing

Chapter II

BACKGROUND

2.1 Quantum Basics

This segment will offer a brief summary of the essentials of quantum computing, mainly based on the textbook by H.Y. Wong. (Wong, 2023).

Quantum Bits

The study of quantum computing has become a cutting-edge discipline that uses the ideas of quantum mechanics to resolve challenging issues that traditional computers are unable to handle. Classical bits, the foundation of traditional computer systems, function in the binary states of 0 and 1. Quantum computers, on the other hand, use quantum bits (qubits), which can exist in a superposition of states and process exponentially more data at once. The basis of quantum computing is this special quality as well as quantum phenomena like entanglement and quantum interference.

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \quad (2.1)$$

where α and β is a complex number with constraints of

$$|\alpha|^2 + |\beta|^2 = 1 \quad (2.2)$$

For this single qubit, the state is $|\psi\rangle$. You can think of $|0\rangle$ and $|1\rangle$ as fancy notations for the classical bits 0 and 1, respectively. One way to think of α and β , which we call amplitude, is as a number that indicates how much 0 or 1 this single qubit is. The probability that the qubit will behave as if it had always been $|0\rangle$ when observed is represented by $|\alpha|^2$, and for $|1\rangle$, by $|\beta|^2$. Using linear algebra, the equation can be better understood by considering $|0\rangle$ and $|1\rangle$ as bases in a two-dimensional complex vector space. Since a single qubit $|\psi\rangle$ is merely a unit vector in the space that the bases span, we can use two complex numbers to describe it.

Typically, we use a collection of bits or qubits rather than a single bit or qubit in our computations. Assume that we have an n -bit register that can be described with n bits of information in the case of classical bits. The register can store 2^n data, which is subsequently encoded to represent a value in our calculation. Let's first examine a system of two qubits in the context of qubits, which can be represented by

$$|\psi\rangle = (\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle) \quad (2.3)$$

$$|\psi\rangle = \alpha_1\alpha_2 |00\rangle + \alpha_1\beta_2 |01\rangle + \alpha_2\beta_1 |10\rangle + \beta_1\beta_2 |11\rangle \quad (2.4)$$

In linear algebraic terms, a quantum system of two qubits can be thought of as a four-dimensional vector space that is spanned by four bases: $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$. Similar to this, the space for a three-qubit system is made up of eight bases: $|000\rangle$, $|001\rangle$, $|010\rangle$, and $|111\rangle$. A system of n qubits can generally be described by

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes |\psi_3\rangle \otimes \cdots \otimes |\psi_n\rangle \quad (2.5)$$

which requires the description of a 2^n complex number. This implies that, in comparison to classical systems, we are able to store a lot more information. Note that even with a large amount of data stored in the system, we can only see n bits of information when we watch the system to extract information. This is the mistake that many people make when they first learn about quantum computing. Although a quantum system may store and process a lot of information at once (this will be covered in a later section), it only tells us n . Therefore, we must extract n bits of information that are relevant to us in order to do *fast computation*.

Superposition and Entanglement

Entanglement and superposition are two important phenomena that make quantum computing possible. Qubits can exist in a combination of $|0\rangle$ and $|1\rangle$ thanks to superposition, which makes computing parallel. The state of one qubit is reliant on the state of another, independent of distance, a property known as entanglement. Information sharing and correlation are made possible by this interconnection, which is not possible with classical systems.

The term "superposition" describes the qubit's ability to be both 0 and 1 at the same time. The set of n independent vectors that span an n -dimensional vector space is referred to as the basis. An infinite set of n vectors can form a basis set for a non-trivial vector space. Therefore, $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$ are just a few of the many bases we can select for a two-qubit system. This collection of bases is known as the computational basis. One qubit may be in a specific state of one basis or in a superposition of another basis. For example,

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \quad (2.6)$$

is a superposition in computational basis but a definite state in $(|0\rangle + |1\rangle)$ and $(|0\rangle - |1\rangle)$ basis. The concept of "entanglement" is another significant occurrence in quantum processing. When two or more particles—in this case, qubits—cannot be explained separately, it's called entanglement. An excellent illustration of an entangled condition is

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle \quad (2.7)$$

This state is entangled because the two-qubit system must be defined as a whole rather than by the states of its individual qubits. Since we cannot express a vector in space with a single basis but rather with a mix of bases, it is simpler to understand from the perspective of linear algebra.

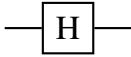
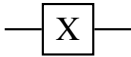
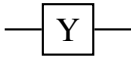
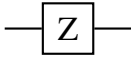
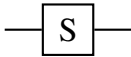
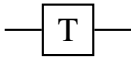
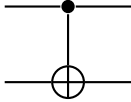
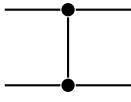
Quantum Computation

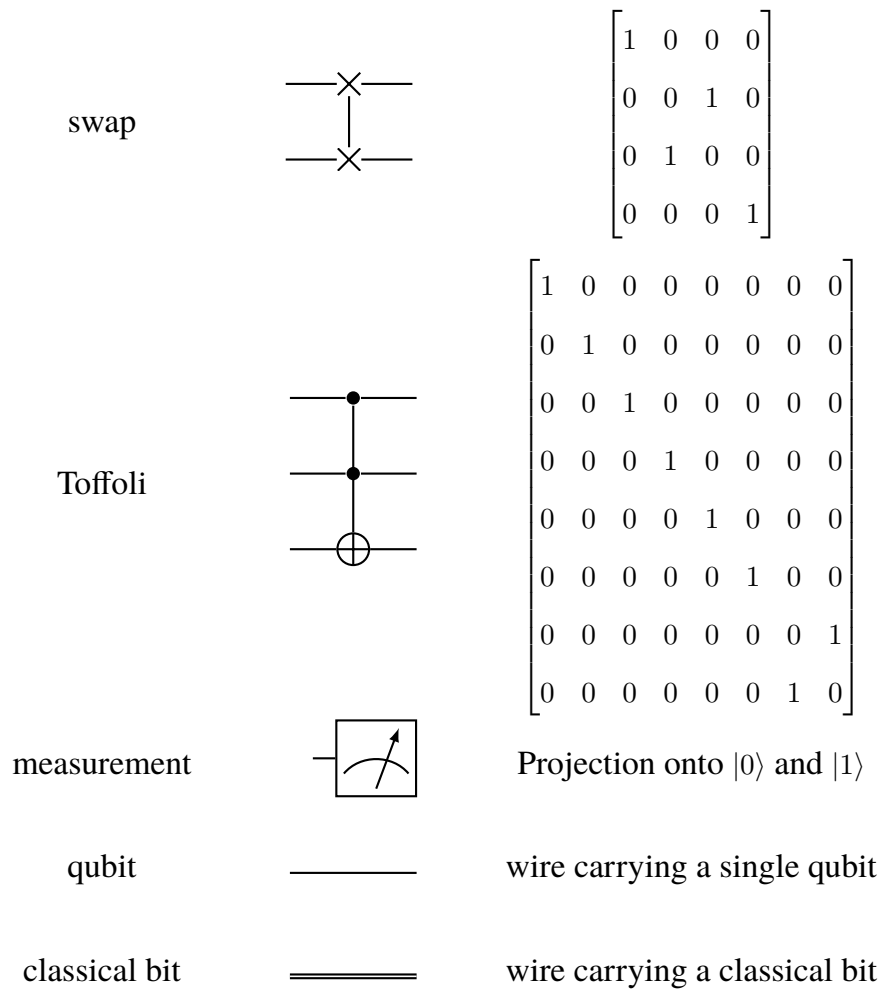
In quantum computing, a quantum gate is an essential part that manipulates a qubit's quantum state by carrying out a unitary operation. Physical gates, such a NAND gate made of transistors, are positioned physically within a circuit to transmit electrical signals in classical computing. The physical transport of qubits or the signals that go with them over space, on the other hand, is usually not a part of quantum computing. To achieve the needed transformations, qubits are instead subjected to external operations like laser or microwave pulses. The evolution of a quantum algorithm, therefore, is not a spatial arrangement but the passage of time. Although

there are few exceptions, such as trapped-ion quantum computers, where ions are transported, qubits often stay stable. Even in these cases, the spatial signal flow of classical systems is not very similar.

Example of quantum gates

The following list includes a few chosen instances of typical gates and circuit symbols used in quantum computing. Each gate is included along with its name, matrix representation, circuit model representation, or a brief description of its function.

Hadamard		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Pauli-X		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Phase		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
controlled-Not		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
controlled-Z		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$



Quantum Algorithm

There are three main categories into which algorithms in quantum computing can be divided.

Quantum Algorithms Based on Quantum Fourier Transform (QFT) - This category encompasses algorithms that exploit the Quantum Fourier Transform, a quantum analogue of the classical discrete Fourier transform, to solve problems related to periodicity and structure. The most notable example is Shor's Algorithm (Shor, 1997), which efficiently factors large integers and calculates discrete logarithms.

These algorithms have profound implications for cryptography, as they can break widely-used encryption protocols such as RSA. QFT-based algorithms are also employed in quantum phase estimation and other applications requiring precision in analyzing frequency components.

Amplitude Amplification Algorithms - These algorithms, like Grover's Algorithm (Grover, 1996), concentrate on optimisation and search issues. In order to suppress wrong solutions and increase the likelihood of desired outcomes, they employ quantum interference. When searching an unstructured database, for example, Grover's approach offers a quadratic speedup, lowering the time complexity from $O(N)$ in traditional systems to $O(\sqrt{N})$. This class of methods is especially helpful for huge dataset problems, machine learning, and optimisation.

Quantum Simulation - Effectively simulating quantum systems, a process that is computationally costly for traditional computers, is one of quantum computing's main advantages. This class of algorithms is made to simulate how quantum mechanical systems—like molecules, materials, and chemical reactions—behave. Scientists may investigate systems that are normally impossible to study classically thanks to these simulations, which have important applications in quantum chemistry, materials science, and drug development.

Chapter III

REVIEW OF LITERATURE

A basic problem in combinatorial optimisation, the Maximum Cut (MaxCut) problem looks for a graph partition that maximises the sum of weights across the partitioned edges. Since this problem is NP-hard, it is best studied using approaches based on Quadratic Unconstrained Binary Optimisation (QUBO), which have become increasingly popular for enabling both quantum and classical methods to effectively solve complicated graph-based problems.

3.1 QUBO and Exact Methods for MaxCut

Numerous approaches have been investigated to effectively solve MaxCut with QUBO, particularly as academics search for solutions to deal with big graph instances. To address the combinatorial difficulty of MaxCut on a smaller scale, the QUBO formulation was frequently combined with exact algorithms in early work, such as branch-and-bound or separation approaches (Häner et al., 2024). By adding penalty terms to the QUBO structure to loosen constraints, more recent research builds upon these foundations and makes the solution space simpler to explore and possibly faster to calculate (Rehfeldt et al., 2022; Fakhimi et al., 2021).

Our work extends these approaches by suggesting adjustments that strike a compromise between computing efficiency and accuracy, even for situations of expansive graphs.

3.2 Quantum Approximate Optimization Algorithms (QAOA) and Quantum Annealing

Among quantum techniques, the Quantum Approximate Optimization Algorithm (QAOA) is one of the most widely applied for solving the MaxCut problem, as it leverages quantum properties like superposition to find near-optimal solutions in large solution spaces. Studies by Farhi et al. (2014) utilized QAOA to handle combinatorial optimization problems, including MaxCut, while achieving a bounded approximation ratio. Additionally, de Falco and Tamascelli (2011) introduced quantum-inspired annealing to further explore quantum potential in solving combinatorial problems with QUBO formulations (Fakhimi et al., 2021; Zhou et al., 2022). However, limitations remain regarding the scalability of QAOA due to current qubit availability, as highlighted in QAOA-in-QAOA approaches that partition large graphs into smaller, manageable subproblems and later merge solutions (Zhou et al., 2023). Our research similarly addresses scalability constraints by introducing a QUBO formulation that can operate efficiently on large graphs without needing extensive quantum resources, thus aligning with but extending beyond traditional QAOA methodologies.

3.3 Hybrid Quantum-Classical and Divide-and-Conquer Approaches

A notable trend in recent research is the hybrid quantum-classical approach, where QUBO-based MaxCut solvers operate within a framework that uses classical methods for preprocessing or subproblem resolution, followed by quantum optimization (Suppakitpaisarn and Hao, 2024). In particular, divide-and-conquer techniques are effective in decomposing large graph instances into subproblems that are solved independently and later reassembled. This strategy has shown promise in large-scale applications by overcoming qubit limitations and enhancing solution quality.

through layered optimization (Zhou et al., 2022).

Our approach takes a similar stance by developing an iterative decomposition model that allows QUBO to perform efficiently on large graphs, even when quantum resources are limited. However, our work distinguishes itself by optimizing the merge process to ensure minimal quality loss between subproblem and final solutions, a gap in current hybrid methods.

3.4 Comparative Analysis and Positioning of Our Research

The primary difference in our approach lies in how we extend QUBO formulations beyond traditional partitioning and quantum-classical hybrids to offer an efficient yet scalable solution for large-scale MaxCut problems. Where earlier QUBO adaptations focused primarily on problem reformulation (Quintero et al., 2021), and quantum-based solutions like QAOA aimed at approximations, our model addresses computational boundaries directly by refining the QUBO framework to handle high-dimensional graphs with reduced quantum resource demands.

In summary, while previous works have laid the groundwork through QAOA, hybrid strategies, and partition-based optimization, our work takes a novel approach in maximizing the efficiency of QUBO for large graphs. This research not only expands on QAOA’s potential in a resource-constrained environment but also enhances the theoretical basis of QUBO for large-scale combinatorial problems, bridging the gap between current capabilities and practical applications. Through these contributions, we aim to advance the efficacy of QUBO-based techniques for solving MaxCut in real-world, large-scale contexts.

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